

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459546.7927 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1519 +/- 0.0081
Transit depth (Rp/Rs)^2	2.31 +/- 0.24 [%]
Orbital Inclination (inc)	90.0 +/- 0.44
Airmass coefficient 1 (a1)	1.282 +/- 0.013
Airmass coefficient 2 (a2)	-0.233 +/- 0.016
Scatter in the residuals of the lightcurve fit is	0.98 %
Best Comparison Star	None
Optimal Aperture	4.53
Optimal Annulus	15.08
Transit Duration (day)	0.10892 +/- 0.00089

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1519 ± 0.0081 vs 0.131 (deviation 2.03σ)
Duration fit vs published	0.10892 d vs 0.1075 d (deviation 1.25σ)
Mid-time O–C	-4.2 min
Depth SNR	14.7
Residual scatter	0.98 %

Light curve

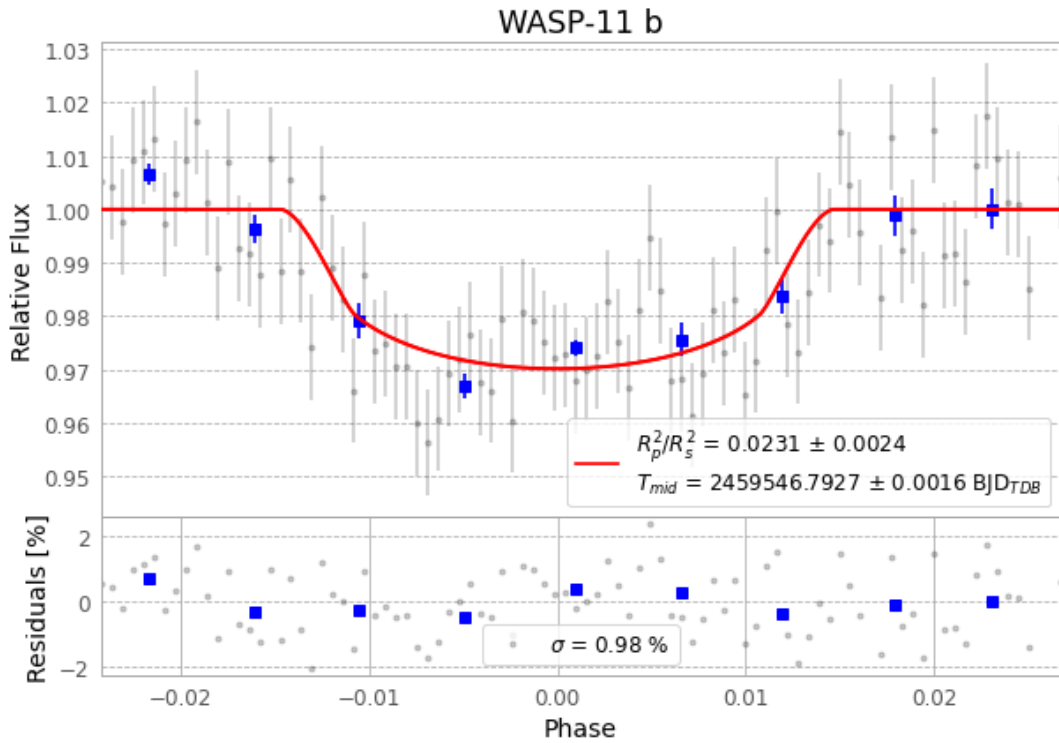


Figure 1 — FinalLightCurve_WASP-11 b_2021-11-27.png (SHA-256 a2f8e2c16ca2bd30...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [415, 247], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 955f471d88ed027d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

90 FITS frames (59 MB), HATP-10211128044212.FITS → HATP-10211128091514.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-211128.log (c34d6715b270...), FinalLightCurve_WASP-11 b_2021-11-27.png (a2f8e2c16ca2...), FinalLightCurve_WASP-11 b_2021-11-27.csv (eb230aab4501...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 19:00Z.